

Exploring Accent Variability, Phrase Difficulty, and Speaker Distance in Robot-Enabled Speech Recognition

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Abstract— This research paper investigates the performance of an open-source Audio Speech Recognition (ASR) model deployed on a robot in diverse linguistic and social contexts. Focusing on accents, phrase difficulty, and speaker distance, the study aims to fill existing gaps in the literature by deploying the ASR model in a social setting, on an embodied robot, and across multiple accent groups. In particular, the research examines potential application of ASR from the lens of educational settings for hearing-impaired individuals. Utilizing the Google Speech Recognition model, the study presents quantitative and qualitative results, revealing challenges faced by users and providing insights into translation accuracy and listener comprehension. Though the findings of the study were inconclusive, we conjecture that if the study were performed on a larger scale, given accented English, increased phrase difficulty, and increased speaker distance, the accuracy of the ASR model's transcription would significantly decrease.

Keywords—Audio Speech Recognition, accent variability, phrase difficulty, speaker distance, embodied robot, Google Speech Recognition, sociolinguistics

I. INTRODUCTION

In recent years, Audio Speech Recognition (ASR) has made massive strides and is a field that continues to grow. However, the effectiveness of ASR models across diverse linguistic factors particularly concerning accent variability, phrase difficulty, and speaker distance, remains underexplored. This study aims to fill this gap. Leveraging an open-source speech recognition model deployed on a robot, we systematically examine its performance nuances with different accents, complexities of phrases, and varying distances of the speaker. This research is particularly significant in the context of aiding individuals with hearing impairments in the classroom, addressing the need for more inclusive educational technology. By analyzing the interplay between the three variables, our study seeks to contribute to the development of more adaptive and universally accessible speech recognition systems in robotic platforms.

II. RELEVANT WORK

Following the rise of voice assistants such as Alexa and Siri in the past few years, there have been well-documented studies of these voice technologies having speech recognition difficulties for accented English speakers, particularly Scottish accents. In one study that analyzes how well ASR models from Amazon, IBM, Google, Microsoft, and Apple, researchers find a 2-12% absolute performance difference between American English and accented English (i.e. British, Indian, French, East Asian, etc.) and a 16-49% absolute performance difference between the various accented English groups^[2]. In another study, after deploying a social robot using 151 Scottish participants, the researchers found that the robot had difficulty with translating English spoken with a Scottish accent^[1]. Though there has been much literature examining the performance of Audio Speech Recognition Models on non-American accents, much of it involves objectively analyzing model performance rather than interactions between a human and embodied robot in a social setting, testing the ASR model on only one accent group, and/or not determining the impact of factors such as phrase difficulty and speaker distance on the model's performance. In this study, we seek to address all of these limitations and further analyze the potential of using an Audio-Speech Recognition robot to support hearing-impaired individuals in a classroom setting.

III. METHODS

This study investigates the performance of Audio Speech Recognition (ASR) models in recognizing speech from English speakers with non-American accents. The focus is on determining how well the Google Speech Recognition model, when integrated into an assistive robot, facilitates communication between an English speaker with a non-American accent and a hearing-impaired individual. This research addresses the notable gap in ASR technology's performance not being analyzed from the perspective of phrase difficulty and speaker volume/distance nor being evaluated on an embodied robot to explore human-robot interaction.

The Google Speech Recognition model was chosen due to its accessibility and compatibility with our assistive robot, Pepper's operating system (NaoQi SDK). We modified

Pepper's speech recognition module to visually indicate its recording (by showing a green screen) and response phases.

Our study included six early-20s undergraduate students, half being male and the other half being female. The group comprised five First-Generation Americans and one non-American, representing a range of ethnic backgrounds: Indo-Guyanese-American (2), Brazilian (1), Indian-American (1), Chinese-American (1), and Haitian-American (1).

Our study was divided into five rounds with two control group rounds testing the American accents of two speakers, followed by three main rounds testing the Guyanese, Chinese, then Brazilian accent. We had three independent variables: Accents, Listener Types, Speech Complexity, Speech Speed. We studied four accent groups; American, Chinese, Guyanese, and Brazilian. Our listeners were hearing-impaired individuals, non-hearing-impaired individuals, and the Pepper robot. We want to disclose that we did not actually have hearing-impaired participants due to a lack of access, so we used noise-canceling headphones playing white noise to simulate being hearing-impaired. In speech complexity, each speaker would say two “easy” phrases followed by two “hard” phrases. Lastly, our speakers in each accent group would do one round of saying the four phrases at a set “close” distance (determined to be three feet away from Pepper) and another round at a “far” distance (six feet away from Pepper). We will also note that the hearing-impaired listeners were at a distance of three feet away from Pepper throughout the study. There were two dependent variables being observed, which consisted of translation accuracy (measuring the Translation Error Rate (TER) of the human transcriber compared to Pepper) as well as the listener comprehension of the hearing-impaired listener.

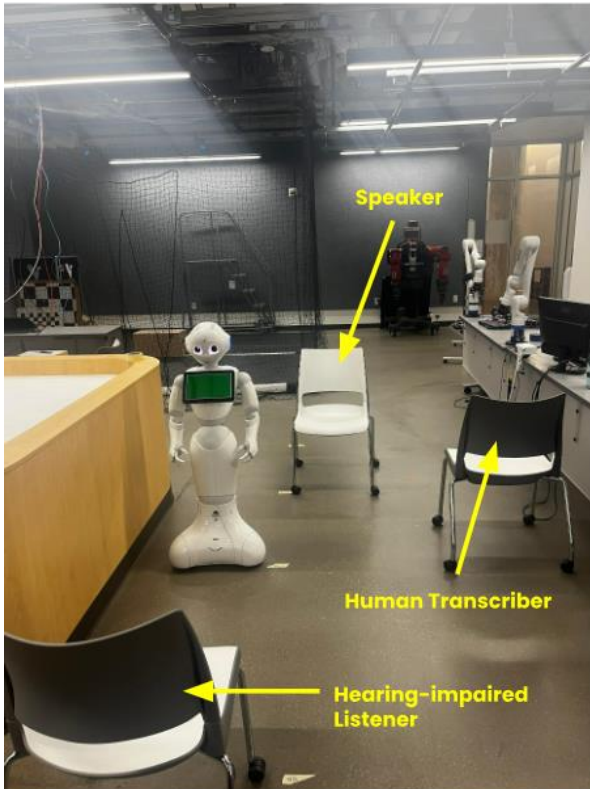


Fig. 1. The experimental setup involved a hearing-impaired listener, human transcriber, speaker, and Pepper robot positioned as pictured.

The procedure (as shown in Figure 2) followed included a series of five rounds, each of which involved one speaker sitting across one hearing-impaired listener with Pepper placed between them. Within each round, we had one human transcriber for the speaker’s close distance, then another for the speaker’s far distance, as pictured in Figure 1. The human transcriber sat in alignment with the robot at some distance to the right. There were two groups: Control Group (testing American accents in two rounds) then the Main Group (testing the three accents in three separate rounds).

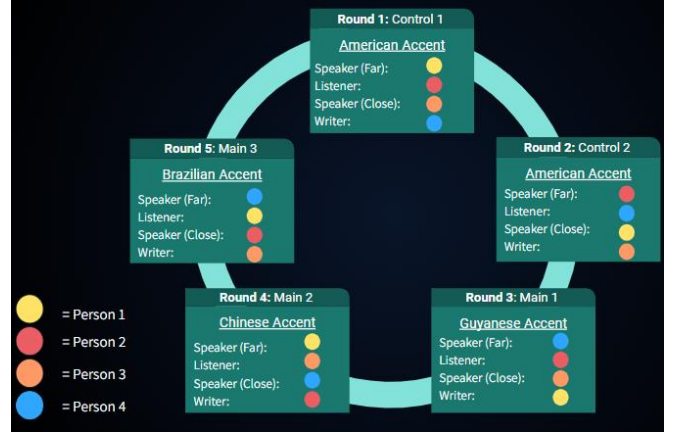


Fig. 2. The experimental procedure consisted of five rounds, with the first two rounds being control rounds that tested the ASR robot on an American accent and the following three rounds being main rounds in which Guyanese, Chinese, and Brazilian accents were tested. Four participants were rotated in the roles of Writer, Speaker (Close, Far), and Listener.

For data collection, Pepper’s screen and command prompt outputs were recorded for speech translation data. Human transcribers used Google Forms to enter their translations, with separate forms for each distance and round.

To analyze the data, we created a Python script to calculate the Translation Error Rate (TER) using the Levenshtein distance metric. The analysis involved comparing human and robot translations at different distances and with varying phrase difficulty, averaging the TERs across phases and rounds.

IV. RESULTS

Figure 3 demonstrates the average human and robot errors for each round when the speaker was positioned 3 feet away from Pepper. In all rounds, the average robot TER was significantly higher than the average human TER. Figure 4 demonstrates the average human and robot errors for each round when the speaker was positioned 6 feet away from Pepper. Once again, for all rounds other than Round 5, the average robot TER was higher than the average human TER. As shown by Figure 5, the human TER was 6.6% and robot TER was 45.8% for the “close” speaker positioning, while the human TER was 15.6% and robot TER was 42.6% for the “far” speaker positioning. Additionally, at the end of the study, each speaker and hearing-impaired listener was asked to fill out a survey comprised of Likert-scale questions. Below are some of the survey results:

Listener-Facing Survey Results

Comprehensibility: On a scale of 1-5, how clear and easy-to-understand was the transcribed text generated by the system? (1 = not easy at all, 5 = very easy)

- 1: 25% (1 responder)
- 2: 50% (2 responders)
- 3: 0% (0 responders)
- 4: 0% (0 responders)
- 5: 25% (1 responder)

Future Use: On a scale of 1-5, how much would you consider recommending this system to a hearing-impaired person? (1 = would not recommend, 5 = would highly recommend)

- 1: 25% (1 responder)
- 2: 25% (1 responder)
- 3: 25% (1 responder)
- 4: 0% (0 responders)
- 5: 25% (1 responder)

Speaker-Facing Survey Results

Ease of use: On a scale of 1-5, rate how easy the system was to use. (1 = not easy at all, 5 = very easy)

- 1: 0% (0 responders)
- 2: 0% (0 responders)
- 3: 25% (1 responder)
- 4: 25% (1 responder)
- 5: 50% (1 responder)

Future Use: On a scale of 1-5, given a scenario where you are asked to communicate with a hearing-impaired individual, how likely would you be to use this system? (1 = not likely at all, 5 = very likely)

- 1: 0% (0 responders)
- 2: 50% (2 responders)
- 3: 0% (0 responders)
- 4: 0% (0 responders)
- 5: 50% (2 responders)

Average Errors for Close Distance (%)		
	Average Human Error	Average Robot Error
Round 1	2.00	22.00
Round 2	4.00	51.00
Round 3	18.00	48.00
Round 4	1.00	83.00
Round 5	8.00	24.00

Fig. 3. Table comparing average human and robot errors at close distance (3 ft) to speaker.

Average Errors for Far Distance (%)		
	Average Human Error	Average Robot Error
Round 1	7.00	54.00
Round 2	23.00	38.00
Round 3	2.00	47.00
Round 4	4.00	51.00
Round 5	42.00	23.00

Fig. 4. Table comparing average human and robot errors at far distance (6 ft) to speaker.

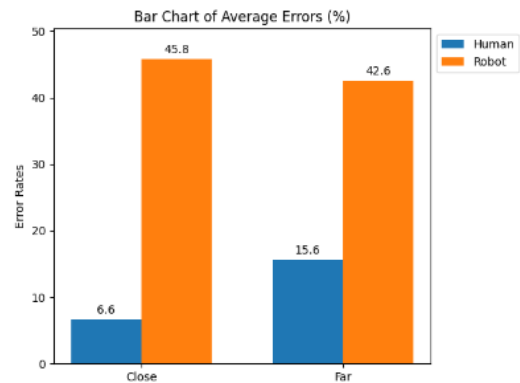


Fig. 5. Bar chart comparing average human and robot errors across all rounds at both close and far distances.

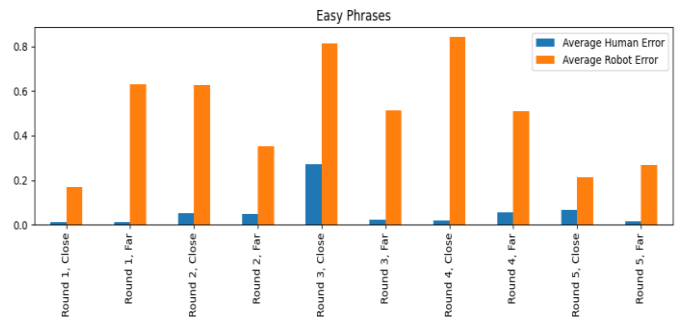


Fig. 6. Bar chart comparing average human and robot errors per round (at close and far distances) for easy phrases.

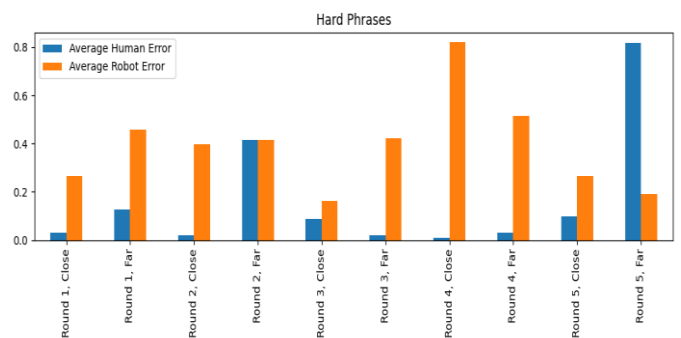


Fig. 7. Bar chart comparing average human and robot errors per round (at close and far distances) for hard phrases.

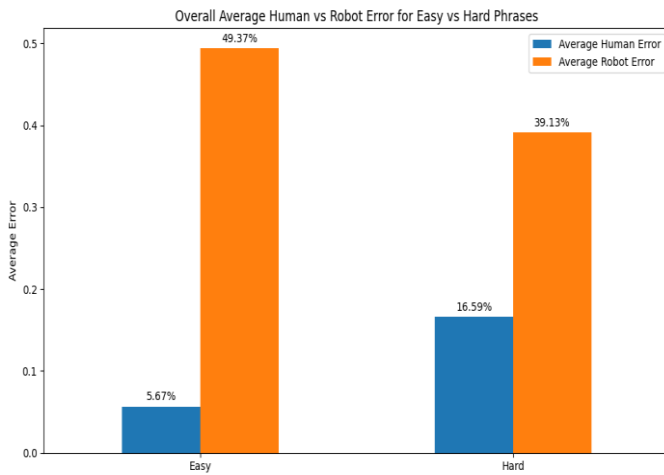


Fig. 8. Bar chart comparing average human and robot errors across all rounds for easy and hard phrases.

Based objectively on our results above, we assessed our original two hypotheses using the statistical t-test:

1. The translation accuracy and listener comprehension levels will be lower for non-American accents compared to American accents ($p < 0.5$). There was no statistically significant difference between the translation accuracies for non-American vs. American accents.
2. ASR performance and listener comprehension will decrease with increased speaker distance and phrase difficulty ($p < 0.5$). There was no statistically significant difference between the translation accuracies for close vs. far speakers and easy vs. difficult phrases.

Therefore, the results of the study were inconclusive.

V. DISCUSSION

In this section, we interpret the results of our study and evaluate the potential for an ASR robot to be used as an educational tool for hearing-impaired users in a classroom.

Accent Variability & Speaker Distance

- At a close distance (3 ft), the average robot error was 36.5% when transcribing English with an American accent. The performance of the robot dropped to a 48% error during Round 3 (Guyanese Accent) and 83% error during Round 4 (Chinese Accent). Unexpectedly, the robot's performance improved during Round 5 (Brazilian Accent), yielding only a 24% error. One reason for this unexpected result was that the Round 5 speaker articulated their phrases much more clearly and loudly than the other speakers, potentially showing that difficulties with translating phrases spoken with a non-American accent can somewhat be combatted with higher volume and signal-to-noise ratio.

- At a far distance (6 ft), the average robot error was 46% when transcribing English with an American accent. The performance of the robot dropped very slightly to 47% error during Round 3 (Guyanese Accent) and a 51% error during Round 4 (Chinese Accent). Once again, the robot's performance improved to a 23% error during Round 5, showing that accent translation and speaker distance barriers can be overcome with a high speech volume. This, however, is not the case with all speakers (especially with lecturers in a classroom) and necessitates a method to increase the signal-to-noise ratio for the input to the ASR model.
- The robot's high error rate for American accent English transcription can be attributed to low speaker volume. If speaker volume were somehow standardized across rounds, there may have been a more apparent negative correlation between robot transcription accuracy and accent variability/speaker distance.

Accent Variability & Phrase Difficulty

- Upon examining the bar charts illustrated in figures 6 and 7, it becomes evident that there is minimal disparity between the errors made by humans when dealing with easy and difficult phrases. A similar pattern emerges in the case of robot translation errors, with little discernible difference between easy and hard phrases. In summary, the human error rate remained relatively low, standing at approximately 5.67% for easy phrases and 16.59% for hard phrases, whereas the robot's error rate was notably higher, reaching approximately 49.37% for easy phrases and 39.13% for hard phrases.
- This null result may have been due to how we determined our easy vs. hard phrases, which can be quite subjective. We used common microphone testing phrases to serve as the "easy" phrases in each of our rounds, since they are meant to be balanced phonetically. Next, we researched words that are both commonly difficult for non-American individuals to pronounce properly in the standard American dialect as well as difficult for Americans to understand, then we incorporated them into phrases which we labeled as being "hard" to be translated/transcribed by both Pepper and a human.
- One example of an easy vs. hard phrase is in round 2 when the speaker was at a far distance. One of the two easy phrases was "Seven crows cawed as the sun climbed the sky", while one of the hard phrases were "His hypothesis was based on a premise so flimsy it could collapse at any moment."
- Our findings suggest that the robot's error rate significantly surpasses that of humans. Consequently, there remains substantial room for improvement in the domain of ASR for embodied robots. Employing a standardized and more objective approach to determine the complexity of

phrases may yield more definitive results in future studies

User Feedback

Following the study, listeners and speakers were asked to comment on their opinions regarding the use of the ASR robot for a hearing-impaired individual in the classroom setting. Though the results of the study were inconclusive, the users' feedback gave us a window of perspective into how the ASR robot could be improved for future iterations of the study. One participant's feedback is as follows:

"As a listener I was hoping to see the translation real time. I was also noticing that sometimes I was naturally trying to read the lips of the person talking instead looking at the robot because I wanted to see the person's facial expression."

Referring to the 1-2 second delay that the robot takes to display the transcription of the speaker's sentence, the user noted that this reaction time made it difficult for the listener to keep up with speaker, which would be impractical in a classroom setting, where lectures are often fast-paced. Additionally, they noted that the robot's transcription did not incorporate grammar or semantics, so they had to rely on the speaker's facial expressions to contextualize what they were saying. Another participant reported the following:

"I think it could do very well and be extremely helpful if it has a little bit more time to train and improve its model... As a speaker I was making sure I was loud and speaking a little more slowly than I usually would."

Though many of the users saw the high potential of the ASR robot, they felt that they had to adapt to the robot's capabilities, as opposed to the robot adapting to their needs. Another user noted that having to repeat their sentence more loudly than they would normally speak was frustrating. In a classroom setting, it is not always possible to ask the lecturer to speak more slowly and loudly for the robot to hear, and it is necessary to build features that allow the robot to filter out background noise and focus solely on the speech that it needs to transcribe.

Based on the survey results above, on average, listeners rated the comprehensibility of the robot's transcription as a 2.5/5 and the likelihood of recommending to a hearing-impaired user as 2.75/5. On average, speakers rated the system's ease-of-use as 4.25/5 and the likelihood of recommending to a speaker communicating with a hearing-impaired user as 3.5/5. Overall, we see that despite the robot's high potential for supporting a hearing-impaired user in an educational setting, limitations such as its inability to filter background noise as speaker volume/distance increase, autocorrect based on context, display translations without a 1-2 second delay, and incorporate semantics into its transcriptions prevent it from being a viable support tool in its current state.

Limitations

- Due to the small scale of the study and limited volunteers, two of the accents (Guyanese and Chinese) were impersonated and not real. Furthermore, each speaker was assumed to represent

an entire accent group, not accounting for dialects or variations in accents within the accent group.

- Due to the small scale of the study, participants were not actually hearing-impaired and wore noise-canceling headphones that played white noise to simulate a hearing impairment.
- The true difficulty of "easy" and "hard" phrases is subjective. In the future, having a metric to quantitatively classify certain phrases as "easy" or "hard" would be more ideal.
- Certain rounds were disproportionately impacted by low speaker volume. The robot could not fully capture phrases spoken at a low volume, especially in Round 4 (Chinese accent group). On the other hand, the speaker in Round 5 (Brazilian accent group) spoke at a very high volume, so their accent, distance, and phrase difficulty did not have a negative impact on the robot's transcription accuracy.
- There was a high level of white noise in the AI MakerSpace area, which may have also impacted the robot's ability to capture phrases that were spoken to it.
- The Google Speech Recognition model lacked features that allowed it to auto-correct its transcription based on the context of the phrases spoken. Therefore, when the word "freeze" was transcribed as "breeze", it significantly altered the meaning of the sentence for the hearing-impaired speaker, to whom the sentence no longer made much sense.

Future Work

In the future, we would like to perform the study on a larger scale, with many more participants representing a wider range of accent groups. Additionally, we would like to deploy more advanced ASR models with improvements such as: increased signal-to-noise ratio, reduced translation delay, and incorporation of punctuation & semantics. Finally, we would like to gain access to and complete a thorough performance comparison between various ASR models to determine which one is the most ideal for Audio Speech Recognition for a hearing-impaired individual in the classroom setting. We believe that an ASR robot that addresses the limitations above would promote equitable use, require low physical effort due to its mobility, allow information to be interpreted in a perceptible way, have a high tolerance for error, and be simple and intuitive to use. All in all, it would enhance the learning experience for hearing-impaired individuals and make the classroom a more inclusive environment.

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